

ANATOMY

QUESTION BANK

SFI MEDICOS KERALA

i. A 23yr old man is brought to the emergency department for evaluation of a traumatic injury to the back of head. The neurological examination revealed tongue deviation to the right, otherwise the cranial nerve exam is normal. What is the most likely location of skull fracture?

- a. Hypoglossal canal
- b. Inferior nuchal line
- c. Jugular foramen
- d. Occipital condyles

✓ Answer: d. Occipital condyles

ii. The chemical used as a preservative in embalming fluid

- a. Formaldehyde
- b. Nitric acid
- c. Glycerin
- d. Thymol

✓ Answer: a. Formaldehyde

iii. The third ventricle is the cavity of

- a. Metencephalon
- b. Myelencephalon
- c. Telencephalon
- d. Diencephalon

✓ Answer: d. Diencephalon

iv. Patella is a

- a. Short bone
- b. Irregular bone
- c. Sesamoid bone
- d. Accessory bone

✓ Answer: c. Sesamoid bone

v. During which stage of development do the three primary germ layers appear

- a. Cleavage
- b. Gastrulation
- c. Neurulation
- d. Blastulation

✓ Answer: b. Gastrulation

vi. Which nerve is affected in this patient (leprosy case with clawing of 4th and 5th fingers)?

- a. Radial nerve
- b. Ulnar nerve

- c. Median nerve
- d. Musculocutaneous nerve

✓ Answer: b. Ulnar nerve

vii. The nerve affected here arises from which cord of brachial plexus

- a. Lateral
- b. Medial
- c. Anterior
- d. Posterior

✓ Answer: b. Medial

viii. What is the deformity mentioned in this case

- a. Ape thumb
- b. Claw hand
- c. Wrist drop
- d. Policeman's tip hand

✓ Answer: b. Claw hand

ix. All the following muscles are supplied by this nerve except

- a. Flexor digitorum profundus
- b. 2nd palmar interossei
- c. Adductor pollicis
- d. 1st lumbrical

✔ Answer: d. 1st lumbrical

x. Which is the autonomous sensory area of this nerve in the hand?

- a. Pulp of middle finger
- b. Pulp of index finger
- c. Pulp of little finger
- d. Pulp of thumb

✔ Answer: c. Pulp of little finger

xi. (A) Muscles of facial expression are supplied by facial nerve

(R) Muscles of facial expression are derived from the second pharyngeal arch mesoderm

✔ Answer: Both A and R are true, and R is the reason for A

xii. (A) Biceps brachii is a strong flexor and supinator of elbow joint

(R) Biceps brachii arises from scapula by two heads

✔ Answer: Both A and R are true, but R is not the reason for A

xiii. (A) Increased intracranial pressure results in optic disc swelling

(R) Optic nerve has an outer meningeal covering

✔ Answer: Both A and R are true, and R is the reason for A

xiv. (A) All lymphoid follicles have germinal centers

(R) A lymph node has several afferent and one efferent lymphatic vessel

✓ Answer: A is incorrect, but R is correct

xv. (A) The dark stained chromatin mass attached to the female interphase nuclear membrane is Barr body

(R) Barr body is derived from one of the two X chromosomes

✓ Answer: Both A and R are true, and R is the reason for A

xvi. The posterior interosseous nerve

i. Passes between the radius and ulna

ii. Lies on the interosseous membrane throughout its course

iii. Is cutaneous to the back of hand

iv. Supplies the extensor carpi radialis longus and extensor carpi radialis brevis

v. It ends in a pseudoganglion

a. i, ii, iii, v

b. i, ii, iv

c. ii, iii, iv

d. i, ii, v

✓ Answer: d. i, ii, v

xvii. The middle meningeal artery

- i. Is a branch of internal carotid artery
- ii. Enters the skull through foramen spinosum
- iii. Gives a branch to circle of Willis
- iv. Divides in the region of pterion
- v. Hemorrhage from the artery is usually subdural

- a. i, ii, iv
- b. ii, iv, v
- c. ii, iv
- d. ii, v

✔ Answer: c. ii, iv

xviii. Hyaline cartilage

- i. Most abundant type of cartilage in the body
- ii. All hyaline cartilages have perichondrium
- iii. Chondrocytes are arranged in clusters
- iv. It is avascular
- v. It has type I collagen fibres which cannot be distinguished from the matrix

- a. i, iii, iv
- b. ii, iv, v
- c. i, ii, iii
- d. i, iv, v

✔ Answer: a. i, iii, iv

xix. Lentiform nucleus

- i. Is related to lateral ventricle
- ii. Consists of putamen and globus pallidus
- iii. Antero inferiorly, caudate and lentiform nuclei are continuous with each other
- iv. Medially it is related to external capsule and laterally to internal capsule
- v. It lies deeply buried in the cerebral hemisphere

a. ii, iv, v

b. ii, iii, v

c. i, ii, iii

d. i, iii, iv

✓ Answer: b. ii, iii, v

xx. Notochord

- i. Arises from the involuting endodermal cells
- ii. Involved in the induction of neural tube
- iii. It gives rise to the vertebral column
- iv. Extends from the prechordal plate to the primitive node
- v. Becomes the appendicular skeleton

a. i, ii, iii

b. ii, iv

c. ii, iii, iv

d. iv, v

✓ Answer: b. ii, iv

i. The vertical group of superficial inguinal lymph nodes is enlarged in

- a. Infection of great toe
- b. Urethritis
- c. Infection of lower back
- d. Infection of lower abdomen

✓ Answer: a. Infection of great toe

ii. Constriction of esophagus is present at all sites except

- a. Cricopharyngeal junction
- b. Where it is crossed by arch of aorta
- c. Where it is surrounded by right crus of diaphragm
- d. Where it is crossed by left principal bronchus

✓ Answer: c. Where it is surrounded by right crus of diaphragm

iii. The coronary sinus

- a. Lies in anterior part of coronary sulcus
- b. Ends in right atrium
- c. Has venae cordis minimae as its tributaries
- d. Develops from right anterior cardinal vein

✓ Answer: b. Ends in right atrium

iv. All are opening into the prostatic urethra except

- a. Prostatic utricle
- b. Ejaculatory ducts
- c. Bulbourethral glands
- d. Prostatic ducts

✓ Answer: c. Bulbourethral glands

v. Artery of hindgut is

- a. Descending artery
- b. Superior mesenteric artery
- c. Inferior mesenteric artery
- d. Coeliac trunk

✓ Answer: c. Inferior mesenteric artery

(vi-x) Case scenario

A 26-year-old woman with two months missed periods complained of acute pain in the lower abdomen.

Ultrasound revealed an enlarged uterus and blood in the pouch of Douglas.

vi. Name the structure that ruptured in the above case

- a. Urinary bladder

- b. Uterus
- c. Uterine tube
- d. Appendix

✓ Answer: c. Uterine tube

vii. The pouch of Douglas is located between

- a. Pubic symphysis and urinary bladder
- b. Urinary bladder and uterus
- c. Uterus and rectum
- d. Rectum and sacrum

✓ Answer: c. Uterus and rectum

viii. The normal site of implantation is

- a. Anterior wall of uterus near the fundus
- b. Posterior wall of uterus near the fundus
- c. Anterior wall of cervix
- d. Posterior wall of cervix

✓ Answer: b. Posterior wall of uterus near the fundus

ix. The operation for female sterilization is

- a. Hysterectomy
- b. Tubectomy
- c. Hysterotomy

d. Colpotomy

✓ Answer: b. Tubectomy

x. Uterine artery is a branch of

a. External iliac artery

b. Internal iliac artery

c. Superior gluteal artery

d. Inferior gluteal artery

✓ Answer: b. Internal iliac artery

(xi–xv) Assertion–Reason

xi. (A) Semitendinosus is grouped under hamstrings

(R) Semitendinosus arises from ischial tuberosity, inserted into tibia and is innervated by tibial component of sciatic nerve

✓ Answer: Both A and R are true, and R is the reason for A

xii. (A) The venesection of the great saphenous vein in emergencies is usually made at the ankle

(R) The saphenous nerve accompanies the great saphenous vein in the leg and foot

✓ Answer: A is incorrect but R is correct

(Venesection is usually done at saphenofemoral junction, not ankle.)

xiii. (A) Diaphragmatic hernia cannot occur through the vena-caval opening

(R) The vena-caval opening is in the central tendon of diaphragm and the wall of the vena cava is firmly attached to the central tendon

✓ Answer: Both A and R are true, and R is the reason for A

xiv. (A) The commonly used anatomical path to reach the coronary arteries for angiography is through the femoral artery

(R) The femoral artery is a continuation of external iliac artery

✓ Answer: Both A and R are true, and R is the reason for A

xv. (A) The common bile duct is blocked in carcinoma head of pancreas leading to jaundice

(R) The common bile duct is embedded in the posterior surface of head of pancreas

✓ Answer: Both A and R are true, and R is the reason for A

xvi. Which of the following is/are true about Fallot's tetralogy

i. Interventricular septal defect

ii. Overriding of aorta

iii. Aortic stenosis

iv. Hypertrophy of right ventricle

a) iii, ii

b) i, ii, iv

c) ii, iii, iv

d) iii, iv

✓ Answer: b. i, ii, iv

xvii. Which of the following is/are true about blood supply of lungs

- i. Pulmonary artery
- ii. Bronchial artery
- iii. Intercostal artery
- iv. Pulmonary vein

- a) iii, iv
- b) ii, iii, iv
- c) i, ii, iv
- d) ii, iii

✓ Answer: c. i, ii, iv

xviii. Which of the following statements is/are true about ovary

- i. The lymphatic drainage is into the para-aortic lymph nodes at the level of first lumbar vertebra
- ii. The ligament of ovary extends from the ovary to the upper end of the lateral wall of uterus
- iii. The ovarian fossa is bounded above by the external iliac vessels
- iv. The obliterated umbilical artery usually lies lateral to the ovary

- a) i, ii, iv
- b) ii, iii, iv
- c) i, ii, iii

d) i, iii

✓ Answer: d. i, iii

xix. Which of the following statements is/are true about boundaries of hepatorenal pouch of Morrison

- i. Inferior layer of coronary ligament
- ii. Inferior surface of right lobe of liver
- iii. Anterior surface of right kidney
- iv. Falciform ligament of liver

a) i, ii, iii

b) i, iii, iv

c) iii, iv

d) i, iii

✓ Answer: a. i, ii, iii

xx. Which of the following statements is/are true about ankle joint

- i. Strengthened by deltoid ligament
- ii. More stable in dorsiflexed position
- iii. Allows inversion and eversion movements
- iv. It is a hinge variety of synovial joint

a) i, ii, iii

b) ii, iii, iv

c) i, ii, iv

d) i, ii, iv

✓ Answer: d. i, ii, iv

Question numbers (i–v) are single response type:

i. Paranasal sinuses opening into middle meatus includes all except:

a) Frontal

b) Anterior ethmoid

c) Middle ethmoid

d) Posterior ethmoid

✓ Answer: d) Posterior ethmoid

ii. The Inferior petrosal sinus drains into

a) Cavernous sinus

b) Internal jugular vein

c) Sigmoid sinus

d) Occipital sinus

✓ Answer: b) Internal jugular vein

iii. The anterior neuropore closes at

a) 24th Day

- b) 25th Day
- c) 27th Day
- d) 28th Day

✓ Answer: a) 24th Day

iv. Role of thymol in embalming fluid:

- a) Antifungal
- b) Antibacterial
- c) Antiviral
- d) Disinfectant

✓ Answer: a) Antifungal

v. Nerve supply of adductor pollicis muscle is:

- a) Deep branch of ulnar nerve
- b) Recurrent branch of median nerve
- c) Superficial branch of radial nerve
- d) Lateral branch of median nerve

✓ Answer: a) Deep branch of ulnar nerve

Case scenario (vi-x)

A 2-year-old child presented with recurrent upper respiratory tract infection, ear ache and mouth breathing. On examination adenoids were noted.

vi. What may be the cause of earache in this child:

- a) Eustachian tube obstruction
- b) Choanal obstruction
- c) Lymphadenopathy
- d) Sinusitis

✓ Answer: a) Eustachian tube obstruction

vii. Which is the most likely involved structure:

- a) Tubal tonsil
- b) Palatine tonsil
- c) Nasopharyngeal tonsil
- d) Lingual tonsil

✓ Answer: c) Nasopharyngeal tonsil (Adenoids)

viii. Which is the nerve related to the palatine tonsil:

- a) Internal laryngeal
- b) External laryngeal
- c) Glossopharyngeal
- d) Hypoglossal

✓ Answer: c) Glossopharyngeal

ix. The lining epithelium of nasopharynx is:

- a) Pseudostratified ciliated columnar
- b) Stratified squamous non-keratinised
- c) Simple ciliated columnar
- d) Simple columnar

✔ Answer: a) Pseudostratified ciliated columnar

x. The muscle passing through sinus of Morgagni:

- a) Levator veli palatini
- b) Tensor veli palatini
- c) Salpingopharyngeus
- d) Stylopharyngeus

✔ Answer: a) Levator veli palatini

Assertion–Reason (xi–xv)

xi.

A: Optic Nerve will not regenerate after injury

R: Optic nerve lacks neurilemmal sheath

✔ Answer: a) Both A and R are true and R is the reason for A

xii.

A: Carcinoma breast can be associated with pain along the medial aspect of arm

R: Enlarged central group of lymph nodes can compress intercostobrachial nerve

✓ Answer: a) Both A and R are true and R is the reason for A

xiii.

A: Human placenta is hemochorial

R: Human placenta is fetomaternal

✓ Answer: b) Both A and R are true and R is NOT the reason for A

xiv.

A: Cerebellar lesions are always ipsilateral

R: Anterior spinocerebellar tract does not cross to opposite side

✓ Answer: c) A is correct but R is incorrect

xv.

A: Metaphysis is the commonest site of osteomyelitis in children

R: Epiphyseal end of diaphysis is called metaphysis

✓ Answer: c) A is correct but R is incorrect

Multiple Response (xvi–xx)

xvi. Parts of limbic system includes:

i. Cingulum

ii. Hippocampus

iii. Pulvinar

iv. Mammillary body

a) i, ii, iii, iv

b) i, ii

c) iii

d) i, ii, iv

✅ Answer: d) i, ii, iv

xvii. Framework made of reticular cells and fibres are seen in:

i. Spleen

ii. Thymus

iii. Tonsil

iv. Lymph node

a) i, ii

b) iii

c) i, ii, iv

d) i, iii, iv

✅ Answer: d) i, iii, iv

xviii. Autosomal recessive inheritance includes:

i. Hemophilia

ii. Sickle cell anaemia

iii. Neurofibromatosis

iv. Thalassemia

a) ii, iii

b) ii, iv

c) i, ii, iii

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✅ Answer: b) ii, iv

xix. Brachialis is supplied by:

i. Radial Nerve

ii. Ulnar Nerve

iii. Median Nerve

iv. Musculocutaneous Nerve

a) i, iv

b) i, iii

c) ii, iii

d) iii, iv

✅ Answer: a) i, iv

xx. Areas devoid of blood brain barrier:

i. Para-central lobule

ii. Pineal gland

iii. Precuneus

iv. Posterior pituitary

a) ii, iii

b) ii, iv

c) i, ii, iii

d) i, ii, iii, iv

✓ Answer: b) ii, iv

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- ii. Hippocampus
- iii. Pulvinar
- iv. Mammillary body

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b) i, ii

c) iii

d) i, ii, iv

✓ Answer: d) i, ii, iv

xvii. Framework made of reticular cells and fibres are seen in:

- i. Spleen
- ii. Thymus
- iii. Tonsil
- iv. Lymph node

a) i, ii

b) iii

c) i, ii, iv

d) i, iii, iv

✔ Answer: d) i, iii, iv

xviii. Autosomal recessive inheritance includes:

- i. Hemophilia
- ii. Sickle cell anaemia
- iii. Neurofibromatosis
- iv. Thalassemia

- a) ii, iii
- b) ii, iv
- c) i, ii, iii
- d) i, iii, iv

✔ Answer: b) ii, iv

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- ii. Ulnar Nerve
- iii. Median Nerve
- iv. Musculocutaneous Nerve

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- b) i, iii
- c) ii, iii
- d) iii, iv

✓ Answer: a) i, iv

xx. Areas devoid of blood brain barrier:

i. Para-central lobule

ii. Pineal gland

iii. Precuneus

iv. Posterior pituitary

a) ii, iii

b) ii, iv

c) i, ii, iii

d) i, ii, iii, iv

✓ Answer: b) ii, iv

Question numbers (i–v) are single response type:

i. Structures passing through lienorenal ligament:

a) Tail of pancreas

b) Short gastric artery

c) Gastroepiploic artery

d) Left gastric artery

✓ Answer: a) Tail of pancreas

ii. The cisterna chyli lies at vertebral level:

- a) T12-L1
- b) T10-T11
- c) T11-T12
- d) L1-L2

✓ Answer: a) T12-L1

iii. The structure passing anterior to scalene tubercle:

- a) Subclavian artery
- b) Lower trunk of brachial plexus
- c) Subclavian vein
- d) Sympathetic chain

✓ Answer: c) Subclavian vein

iv. Adrenal medulla develops from:

- a) Notochord
- b) Neural crest
- c) Splanchnopleuric mesoderm
- d) Somatopleuric mesoderm

✓ Answer: b) Neural crest

v. Paneth cells are seen in:

- a) Stomach
- b) Appendix

c) Large intestine

d) Small intestine

✓ Answer: d) Small intestine

Case scenario (vi-x):

An 18-year-old boy sustained an injury to his right knee following a road-traffic accident. On examination, soft tissue swelling and tenderness over the right knee were noted.

vi. Positive anterior drawer test indicates injury of which ligament:

a) Oblique popliteal

b) Arcuate

c) Anterior cruciate

d) Posterior cruciate

✓ Answer: c) Anterior cruciate

vii. Oblique popliteal ligament is an expansion of:

a) Semitendinosus

b) Semimembranosus

c) Adductor magnus

d) Adductor longus

✓ Answer: b) Semimembranosus

viii. Unhappy triad of knee includes all except:

- a) Anterior cruciate ligament
- b) Medial meniscus
- c) Medial collateral ligament
- d) Posterior cruciate ligament

✓ Answer: d) Posterior cruciate ligament

ix. When the foot is on the ground, medial rotation of femur on tibia is:

- a) Unlocking
- b) Locking
- c) Medial rotation
- d) Lateral rotation

✓ Answer: b) Locking

x. Anserine bursa is related to:

- a) Semitendinosus
- b) Semimembranosus
- c) Biceps femoris
- d) Fibular collateral ligament

✓ Answer: a) Semitendinosus

Assertion–Reason (xi–xv):

xi.

A: Greater omentum is known as policeman of abdomen

R: Greater omentum forms a partition between supracolic and infracolic compartments

✔ Answer: c) A is correct but R is incorrect

xii.

A: Adductor magnus is a true hamstring

R: Ischial part of adductor magnus is innervated by sciatic nerve

✔ Answer: a) Both A and R are true and R is the reason for A

xiii.

A: Direct inguinal hernia protrudes through Hesselbach's triangle

R: Weakness of conjoint tendon contributes to direct inguinal hernia

✔ Answer: a) Both A and R are true and R is the reason for A

xiv.

A: Transverse sinus lies between arterial and venous ends of developing heart tube

R: Transverse pericardial sinus is developed by the degeneration of dorsal mesocardium

✔ Answer: a) Both A and R are true and R is the reason for A

xv.

A: Prostatic venous plexus lies between the true and false capsules

R: During prostatectomy, prostate is removed along with false capsule

✓ Answer: a) Both A and R are true and R is the reason for A

Multiple Response (xvi–xx):

xvi. Structures crossing root of mesentery includes:

i. Right ureter

ii. Left ureter

iii. 3rd part of duodenum

iv. Superior mesenteric artery

a) i, iii, iv

b) iii

c) i, ii, iii

d) i, ii, iv

✓ Answer: a) i, iii, iv

xvii. Stages of lung development include:

i. Pseudoglandular

ii. Canalicular

iii. Glandular

iv. Pseudosaccular

a) iii, ii, iv

b) i, ii

c) i, ii, iv

d) i, iii

✔ Corrected Answer: c) i, ii, iv

xviii. Structures passing through major openings of diaphragm at the level of T12:

i. Aorta

ii. Esophagus

iii. Azygos vein

iv. Thoracic duct

a) i, iii

b) ii, iii, iv

c) i, ii, iii

d) i, iii, iv

✔ Answer: d) i, iii, iv

xix. Structures related to ventral surface of ala of sacrum:

i. Lumbosacral trunk

ii. Inferior gluteal artery

iii. Median sacral artery

iv. Obturator nerve

- a) i, iv
- b) i, ii, iii
- c) ii, iii
- d) ii, iv

✔ Answer: a) i, iv

xx. Muscles attached to perineal body include:

- i. Ischiocavernosus
- ii. Superficial transverse perinei
- iii. Deep transverse perinei
- iv. Sphincter urethra

- a) ii, iii
- b) i, iv
- c) i, ii, iii
- d) ii, iii, iv

✔ Answer: d) ii, iii, iv

Case Scenario (Q1–Q5)

A 45-year-old obese woman presented to OP with severe pain in her right upper quadrant of abdomen after a fatty meal. There was associated nausea, vomiting and radiation of pain to right shoulder.

1. Name the organ involved in this case

- a. Pancreas
- b. Gall bladder
- c. Duodenum
- d. None of the above

✓ Answer: b. Gall bladder

2. What is the clinical diagnosis

- a. Peptic ulcer
- b. Pancreatitis
- c. Cholecystitis
- d. Appendicitis

✓ Answer: c. Cholecystitis

3. Irritation of which nerve caused right shoulder pain

- a. Right gastric nerve
- b. Right phrenic nerve
- c. Left gastric nerve
- d. Right vagus nerve

✓ Answer: b. Right phrenic nerve

4. Mention the diagnostic test most useful in the above condition

- a. MRI
- b. Endoscopy

c. CT Scan

d. Ultrasound

✓ Answer: d. Ultrasound

5. Give the source of development of the involved organ

a. Ventral pancreatic bud

b. Hepatic bud

c. Caecal diverticulum

d. Dorsal pancreatic bud

✓ Answer: b. Hepatic bud

Assertion–Reason (Q6–Q10)

6.

A: Carcinoma prostate can cause metastasis in the vertebral column.

R: Few veins from the prostate pass backwards and drain into internal vertebral venous plexus via veins of Batson.

✓ Answer: Both A and R are True and R is the correct explanation of A

7.

A: External haemorrhoids are painless.

R: The area below the pectinate line is supplied by somatic nerves.

✓ Answer: A is False and R is True

8.

A: Carcinoma of the head of the pancreas commonly presents with obstructive jaundice.

R: The common bile duct passes through a groove in the posterior aspect of the head of the pancreas.

✓ Answer: Both A and R are True and R is the correct explanation of A

9.

A: McBurney's Point is the area of maximum tenderness in acute appendicitis.

R: Posterior relations of retrocecal appendix are the psoas major and iliacus muscles.

✓ Answer: Both A and R are True but R is not the correct explanation of A

10.

A: Portal hypertension can lead to the development of esophageal varices.

R: The left gastric vein forms portocaval anastomosis by connecting the portal venous system with esophageal veins of the systemic circulation.

✓ Answer: Both A and R are True and R is the correct explanation of A

Multiple Response (Q11–Q15)

11. The deep inguinal lymph nodes receive lymph from

- i. Superficial inguinal lymph nodes
- ii. External iliac nodes

iii. Popliteal nodes

iv. Paraaortic nodes

a. i, ii, iii

b. ii and iii

c. i and iii

d. i, iii, iv

✓ Answer: c. i and iii

12. What are the contents of superficial perineal pouch?

i. Bulbospongiosus

ii. Crus of penis/clitoris

iii. External urethral sphincter

iv. Deep transverse perineal

a. i and ii

b. ii, iii, iv

c. ii and iv

d. i, iii, iv

✓ Answer: a. i and ii

13. Which of the following peritoneal ligaments form the boundaries of lesser sac?

i. Greater omentum

ii. Lesser omentum

iii. Gastrosplenic ligament

iv. Lienorenal ligament

a. i, ii, iii, iv

b. i and ii

c. i, iii and iv

d. ii and iii

✓ Answer: a. i, ii, iii, iv

14. Lienorenal ligament contains

i. Left gastroepiploic artery

ii. Splenic artery

iii. Tail of pancreas

iv. Sustentaculum lienis

a. i and ii

b. ii and iii

c. i, ii, iii

d. i, iii, iv

✓ Answer: b. ii and iii

15. All are components of interatrial septum

i. Septum primum

ii. Septum secundum

iii. Septum spurium

iv. Right venous valve

a. i, ii, iii, iv

b. i, ii and iii

c. i, ii and iv

d. i and ii

✓ Answer: b. i, ii and iii

Single Response (Q16–Q20)

16. Which among the following is the short, narrow and least dilatable part of male urethra?

a. Pre-prostatic part

b. Prostatic part

c. Membranous urethra

d. Penile part

✓ Answer: c. Membranous urethra

17. All are contents of the rectus sheath except

a. Rectus abdominis

b. Upper 5 thoracic nerves

c. Superior epigastric vessels

d. Pyramidalis

✓ Answer: b. Upper 5 thoracic nerves

18. Which nerve of the lumbar plexus supplies the cremaster muscle?

- a. Ilioinguinal nerve
- b. Femoral nerve
- c. Genitofemoral nerve
- d. Obturator nerve

✓ Answer: c. Genitofemoral nerve

19. The duodenum develops from which embryological structure?

- a. Foregut only
- b. Midgut only
- c. Hindgut
- d. Both foregut and midgut

✓ Answer: d. Both foregut and midgut

20. The sympathetic nerve supply of the ureter arises from which spinal segments?

- a. T10–T11
- b. T6–T9
- c. T12–L1
- d. L2–L3

✓ Answer: a. T10–T11

Case Scenario (Q1–Q5)

A 44-year-old male patient was brought to emergency department with history of sudden onset of dizziness, difficulty in walking and swallowing. Examination revealed loss of pain and temperature sensations over face, nystagmus, Horner's syndrome and absent gag reflex on right side. Pain and temperature sensations were absent on left side of body.

1. Name the probable clinical condition

- a. Pontine haemorrhage
- b. Wallenberg syndrome
- c. Cerebellar infarct
- d. Millard Gubler syndrome

✓ Answer: b. Wallenberg syndrome (Lateral medullary syndrome)

2. Name the artery affected

- a. Basilar
- b. Anterior inferior cerebellar
- c. Vertebral
- d. Posterior inferior cerebellar

✓ Answer: d. Posterior inferior cerebellar (PICA)

3. Involvement of which cranial nerve nucleus is responsible for difficulty in swallowing

- a. Facial nerve nucleus
- b. Nucleus of tractus solitarius
- c. Nucleus ambiguus

d. Dorsal nucleus of Vagus

✓ Answer: c. Nucleus ambiguus

4. Which of the following tracts are affected here

a. Medial longitudinal fasciculus

b. Lateral spinothalamic tract

c. Medial lemniscus

d. Trigeminal lemniscus

✓ Answer: b. Lateral spinothalamic tract

5. Which functional column is placed medial most in ventral part of brainstem

a. General somatic efferent

b. General visceral efferent

c. Special somatic afferent

d. General somatic afferent

✓ Answer: a. General somatic efferent

Assertion–Reason (Q6–Q10)

6.

A: Air embolism can occur in injuries over posterior triangle of neck.

R: The external jugular vein is adherent to deep fascia where it pierces it to become deep.

✓ Answer: Both A and R are True and R is the correct explanation of A

7.

A: Transitional epithelium is found lining the uterus and helps in its distension.

R: Tight junctions present between surface cells of transitional epithelium prevent separation of cells while allowing stretching.

✓ Answer: A is False and R is True

8.

A: Colour blindness is more common in males than in females.

R: It is inherited as an X-linked dominant trait.

✓ Answer: A is True and R is False

9.

A: Patients with Carcinoma breast can complain of pain along the medial side of arm.

R: Medial cutaneous nerve of arm passes through central group of axillary lymph nodes.

✓ Answer: A is True and R is False

10.

A: Abduction of shoulder joint from 90° to 120° can occur only if the humerus is rotated laterally.

R: Infraspinatus and Teres minor rotate the humerus medially.

✓ Answer: A is True and R is False

Multiple Response (Q11–Q15)

11. The muscles attached to the base of the proximal phalanx of the thumb are –

- i. 1st palmar interosseus
- ii. Opponens pollicis
- iii. 1st dorsal interosseus
- iv. Adductor pollicis

- a. i and ii
- b. i and iii
- c. i and iv
- d. iii and iv

✓ Answer: d. iii and iv

12. Which of the following statements regarding the pharynx are TRUE?

- i. The pharynx extends from the base of the skull to the level of the fifth cervical vertebra.
- ii. Inferior constrictor muscle forms the lower part of the pharyngeal wall.
- iii. The Eustachian tube opens into the medial wall of the nasopharynx.
- iv. The nasopharynx is lined by pseudostratified ciliated columnar epithelium.

- a. i & ii are correct
- b. ii & iii are correct
- c. i & iii are correct
- d. ii & iv are correct

✓ Answer: a. i & ii are correct

13. The proximal radioulnar joint:

- i. Is a pivot joint
- ii. Allows pronation and supination
- iii. Contains an articular disc
- iv. Has the annular ligament around the radial head

a. i, ii, iii, iv are correct

b. i, ii, iii are correct

c. i, ii & iv are correct

d. i & ii are correct

✓ Answer: c. i, ii & iv are correct

14. Which of the following are intrinsic muscles of larynx

- i. Thyroarytenoid
- ii. Cricothyroid
- iii. Sternothyroid
- iv. Stylopharyngeus

a. i, ii, iii, iv are correct

b. i, ii, iii are correct

c. i, ii & iv are correct

d. i & ii are correct

✓ Answer: d. i & ii (Thyroarytenoid, Cricothyroid)

15. What are the components of secondary villi of placenta

- i. Syncytiotrophoblast
- ii. Cytotrophoblast
- iii. Fetal capillaries
- iv. Extraembryonic mesoderm

a. i, ii, iii, iv are correct

b. i, ii, iii are correct

c. i, ii & iv are correct

d. i & ii are correct

✓ Answer: c. i, ii & iv

Single Response (Q16–Q20)

16. All of the following are derivatives of neural crest EXCEPT

- a. Adrenal medulla
- b. Schwann cells
- c. Skeletal muscles
- d. Melanocytes

✓ Answer: c. Skeletal muscles

17. The sinus present in the free border of falx cerebri is

- a. Superior sagittal sinus
- b. Inferior sagittal sinus
- c. Straight sinus
- d. Inferior petrosal sinus

✓ Answer: b. Inferior sagittal sinus

18. Which among the following pierces clavipectoral fascia

- a. Medial pectoral nerve
- b. Lateral pectoral nerve
- c. Basilic vein
- d. Thyrocervical trunk

✓ Answer: b. Lateral pectoral nerve

19. All of the following are examples of white fibrocartilage except

- a. Intervertebral disc
- b. Articular cartilage of temporomandibular joint
- c. Cartilage of pinna of ear
- d. Acetabular labrum

✓ Answer: c. Cartilage of pinna of ear

20. Which process ensures that each gamete receives only one allele of each gene?

- a. Replication

- b. Mutation
- c. Mitosis
- d. Meiosis

✓ Answer: d. Meiosis

Case Scenario (Q1–Q5)

A 6 yr old boy complained of severe ear pain 2 days after he underwent tonsillectomy (removal of palatine tonsils) for chronic tonsillitis. He also had history of severe bleeding during tonsillectomy

1. Irritation of the following nerve can be the cause of ear pain after tonsillectomy

- a. Posterior auricular nerve
- b. Auricular branch of Vagus nerve
- c. Glossopharyngeal nerve
- d. Auriculotemporal nerve

✓ Answer: c. Glossopharyngeal nerve

2. Most common cause of bleeding during tonsillectomy is injury to

- a. Paratonsillar vein
- b. Supratonsillar vein
- c. Infratonsillar vein
- d. Internal jugular vein

✓ Answer: a. Paratonsillar vein

3. The muscle which forms tonsillar bed is

- a. Palatoglossus
- b. Palatopharyngeus
- c. Superior constrictor
- d. Levator palatini

✔ Answer: c. Superior constrictor

4. All the following arteries supply tonsil EXCEPT

- a. Facial artery
- b. Lingual artery
- c. Ascending pharyngeal artery
- d. Posterior auricular artery

✔ Answer: d. Posterior auricular artery

5. Palatine tonsils develop from

- a. Second pharyngeal arch
- b. Second pharyngeal pouch
- c. Third pharyngeal arch
- d. Third pharyngeal pouch

✔ Answer: b. Second pharyngeal pouch

Assertion–Reason (Q6–Q10)

6.

A: Wounds of scalp bleed profusely and heal quickly

R: Scalp is richly supplied with blood vessels and they are unable to retract in injuries as their walls are firmly adherent to fibrous network.

✓ Answer: Both A and R are True and R is the correct explanation of A

7.

A: Pontine haemorrhage can result in pinpoint pupil

R: Parasympathetic part of oculomotor nucleus is involved in pontine haemorrhage

✓ Answer: A is True and R is False

8.

A: Inferior oblique is the sole depressor of adducted eye

R: The optical axis gets aligned to the axis of oblique muscles in adduction

✓ Answer: Both A and R are True and R is the correct explanation of A

9.

A: Pituitary tumours can present with bitemporal hemianopia

R: The optic chiasma contains fibres from temporal part of retina

✓ Answer: A is True and R is False

10.

A: Posterior part of calcarine sulcus is classified as a complete sulcus

R: Posterior part of calcarine sulcus forms calcar avis in posterior horn of lateral ventricle

✅ Answer: A is False and R is True

Multiple Response (Q11–Q15)

11. Which of the following statements about maxillary sinus are true

- i. It is the largest paranasal sinus
- ii. It opens into superior meatus of nose
- iii. It is the first paranasal sinus to develop
- iv. The floor of the sinus is formed by alveolar process of maxilla

- a. i, ii, iii, iv
- b. i, ii and iii
- c. i and iii
- d. i, iii and iv

✅ Answer: d. i, iii and iv

12. Anterior division of mandibular nerve supplies

- i. Tensor tympani
- ii. Mylohyoid
- iii. Masseter
- iv. Lateral pterygoid

- a. i and ii
- b. ii, iii and iv
- c. iii and iv
- d. ii, iii and iv

✔ Answer: c. iii and iv

13. Paralysis of muscles of soft palate can result in

- i. Drooling of saliva
- ii. Nasal regurgitation during swallowing
- iii. Nasal voice
- iv. Defective gag reflex

- a. i, ii and iii
- b. ii and iii
- c. ii, iii and iv
- d. i and iii

✔ Answer: c. ii, iii and iv

14. Identify the correct statements regarding chorda tympani

- i. It is a branch of seventh cranial nerve
- ii. It carries secretomotor fibres from inferior salivatory nucleus
- iii. It is closely related to the medial wall of middle ear
- iv. It carries taste sensations from the anterior two thirds of tongue

- a. i and iv
- b. ii and iii
- c. ii and iv
- d. i, ii, iii and iv

✔ Answer: a. i and iv

15. The apex of thyroid gland is related to

- i. Sternothyroid muscle
- ii. Inferior constrictor muscle
- iii. Middle constrictor muscle
- iv. External laryngeal nerve

- a. i and iii
- b. i and ii
- c. iii and iv
- d. i, ii and iv

✔ Answer: b. i and ii

Single Response (Q16–Q20)

16. Structures that form boundary of lateral ventricles include all EXCEPT

- a. Forceps minor
- b. Forceps major

- c. Tapetum
- d. Optic radiation

✓ Answer: d. Optic radiation

17. The maternal infection which can result in congenital cataract in the fetus

- a. Cytomegalovirus infection
- b. Toxoplasmosis
- c. Chickenpox
- d. Rubella

✓ Answer: d. Rubella

18. Which layer of cerebrum has large pyramidal cells which project to spinal cord

- a. External pyramidal layer
- b. External granular layer
- c. Internal pyramidal layer
- d. Multiform layer

✓ Answer: c. Internal pyramidal layer

19. The approximate vertebral level of T10 spinal segment is

- a. T7
- b. T8
- c. T9
- d. T10

✓ Answer: a. T7

20. The type of joint formed between basi-occiput and basi-sphenoid is

- a. Primary cartilaginous
- b. Secondary cartilaginous
- c. Membranous
- d. Synovial

✓ Answer: b. Secondary cartilaginous

Single Response Type (i–v)

i. AV node is most commonly supplied by

- a) Anterior interventricular artery
- b) Posterior interventricular artery
- c) Conus artery
- d) Circumflex artery

✓ Answer: b) Posterior interventricular artery

ii. Abduction of shoulder is produced by all the following muscles except

- a) Supraspinatus
- b) Deltoid
- c) Teres major
- d) Serratus anterior

✓ Answer: c) Teres major

iii. Which of the following is the mature bone cell?

- a) Chondrocyte
- b) Osteoblast
- c) Osteoclast
- d) Osteocyte

✓ Answer: d) Osteocyte

iv. The posterior smooth part of right atrium develops from

- a) Primitive atrium
- b) Bulbus cordis
- c) Sinus venosus
- d) Truncus arteriosus

✓ Answer: c) Sinus venosus

v. Which of the following type of joint permits only two degrees of freedom of movement?

- a) Hinge
- b) Pivot
- c) Ball and socket
- d) Ellipsoid

✓ Answer: d) Ellipsoid (condyloid joint)

Case Scenario (vi-x)

A 24 year old football player had twisting injury to right knee, pain & swelling along medial aspect.

vi. Which is the most probable structure injured?

- a) Ligamentum patellae
- b) Posterior cruciate ligament
- c) Medial collateral ligament
- d) Lateral collateral ligament

✓ Answer: c) Medial collateral ligament

vii. Which clinical finding(s) will confirm injury to the above said structure?

- a) Tenderness at the medial aspect of knee joint
- b) Positive valgus stress test
- c) Lachmann's test
- d) Both A & B

✓ Answer: d) Both A & B

viii. Which intracapsular structure(s) can be injured along with the structure in the first question?

- a) Anterior cruciate ligament
- b) Posterior cruciate ligament
- c) Lateral meniscus

d) Both B and C

✓ Answer: a) Anterior cruciate ligament

ix. What is the histology of menisci?

- a) Dense regular connective tissue
- b) Fibrocartilage
- c) Hyaline cartilage
- d) Dense irregular connective tissue

✓ Answer: b) Fibrocartilage

x. All the following statements are true EXCEPT

- a) Anterior cruciate ligament is attached to medial condyle of femur
- b) Oblique popliteal ligament is an expansion of semimembranosus tendon
- c) Popliteus muscle prevents the lateral meniscus getting pinched between femur and tibia during flexion
- d) Clergyman's knee affects infrapatellar bursa

✓ Answer: a) Anterior cruciate ligament is attached to medial condyle of femur

Assertion-Reason (xi-xv)

xi.

- A. An accidentally inhaled object will most likely enter the right principal bronchus
- R. It is wider, shorter and more vertical than the left bronchus

✓ Answer: a) Both A and R are true and R is the reason for A

xii.

A. Loss of sensation of the skin over deltoid can occur in fracture surgical neck of humerus

R. Radial nerve winds round the surgical neck of humerus

✓ Answer: c) A is correct but R is incorrect

xiii.

A. Posterior interosseous nerve injury causes finger drop

R. All the extensor muscles of forearm are supplied by posterior interosseous nerve

✓ Answer: a) Both A and R are true and R is the reason for A

xiv.

A. Primitive streak is called as the primary organizer

R. Primitive streak induces the formation of notochord and intraembryonic mesoderm

✓ Answer: a) Both A and R are true and R is the reason for A

xv.

A. Paralysis of muscles of anterior compartment of leg causes foot drop

R. Fracture neck of fibula causes injury to common peroneal nerve

✓ Answer: a) Both A and R are true and R is the reason for A

Multiple Response (xvi–xx)

xvi. Which among the following statements are true regarding internal thoracic artery?

- i) It is a branch of second part of subclavian artery
- ii) It divides at the level of sixth intercostal space
- iii) Terminal branches are superior epigastric and musculophrenic arteries
- iv) It provides lateral mammary branches

- a) i, ii
- b) i, iii
- c) ii, iv
- d) ii, iii

✓ Answer: d) ii, iii

xvii. Which among the following statements are true regarding anatomical snuff box?

- i) Lateral boundary is formed by abductor pollicis longus and extensor pollicis longus
- ii) Cephalic vein occupies its roof
- iii) Floor consists of lunate, scaphoid and base of first metacarpal
- iv) Radial artery is its content

- a) ii, iv
- b) i, iii, iv
- c) i, ii
- d) i, ii, iii, iv

✓ Answer: a) ii, iv

xviii. Which of the following are derivatives of neural crest cells?

- i) Adrenal medulla
- ii) Neurons of sympathetic ganglia
- iii) Lens of eye
- iv) Melanoblasts

- a) i, ii, iii
- b) i, iv
- c) i, ii, iv
- d) i, ii, iii, iv

✓ Answer: c) i, ii, iv

xix. Which among the following statements are true regarding bronchopulmonary segments?

- i) It is a portion of lungs aerated by a terminal bronchiole
- ii) Pyramidal in shape with apex directed towards the surface
- iii) Apical segment of right lower lobe is the most commonly involved lobe in aspiration pneumonia
- iv) Pulmonary arteries are segmental and pulmonary veins are intersegmental in distribution

- a) i, ii
- b) ii, iii
- c) ii, iv

d) i, iii

✓ Answer: c) ii, iv

xx. Which of the following muscles are supplied by the anterior interosseous nerve?

i) Medial half of flexor digitorum profundus

ii) Flexor pollicis longus

iii) Pronator quadratus

iv) Flexor digitorum superficialis

a) i, ii, iii

b) ii, iii

c) i, ii

d) ii, iv

✓ Answer: a) i, ii, iii

Question numbers (i–v) are single response type questions.

i. Which of the following structure develops from fourth pharyngeal pouch?

a) Inferior parathyroid gland

b) Thymus

c) Tonsil

d) Superior parathyroid gland

✓ Answer: d) Superior parathyroid gland

ii. Which of the following nerve plexus is present in the submucosa of gastrointestinal tract?

- a) There is no plexus in submucosa
- b) Coeliac plexus
- c) Auerbach's plexus
- d) Meissner's plexus

✓ Answer: d) Meissner's plexus

iii. Which of the following epithelium lines the distal part of vas deferens?

- a) Stratified squamous non-keratinized
- b) Pseudostratified columnar without stereocilia
- c) Pseudostratified columnar with stereocilia
- d) Simple columnar

✓ Answer: c) Pseudostratified columnar with stereocilia

iv. Which of the following is NOT a content of pudendal canal?

- a) Pudendal nerve
- b) Nerve to obturator internus
- c) Internal pudendal artery
- d) Internal pudendal vein

✓ Answer: b) Nerve to obturator internus

v. What type of cells are type II pneumocytes?

- a) Simple columnar
- b) Stratified columnar
- c) Simple cuboidal
- d) Simple squamous

✓ Answer: c) Simple cuboidal

Question numbers (vi-x) are case scenario based questions:

A 35-year-old male presents to the emergency department with severe pain in the left upper quadrant of abdomen and referred pain in the left shoulder following a car accident. Imaging confirms splenic rupture.

vi. What is the mechanism behind pain referred to the left shoulder in splenic rupture?

- a) Direct injury to the phrenic nerve
- b) Irritation of the left hemidiaphragm by blood in the peritoneal cavity
- c) Compression of brachial plexus
- d) Fracture of clavicle

✓ Answer: b) Irritation of the left hemidiaphragm by blood in the peritoneal cavity

vii. Which cervical nerve roots are involved in transmitting pain to the shoulder?

- a) C1–C2
- b) C3–C5
- c) T1–T3
- d) L1–L3

✓ Answer: b) C3–C5

viii. The main artery which provides blood supply to the spleen is a branch of?

- a) Superior mesenteric artery
- b) Gastric artery
- c) Gastroduodenal artery
- d) Coeliac trunk

✓ Answer: d) Coeliac trunk

ix. Which of the following structure is NOT attached to spleen by ligaments?

- a) Stomach
- b) Left kidney
- c) Left colic flexure
- d) Diaphragm

✓ Answer: d) Diaphragm

x. What is the primary function of the white pulp in the spleen?

- a) Storage of platelets
- b) Immune response
- c) Filtration of blood
- d) Production of red blood cells

✓ Answer: b) Immune response

Question numbers (xi–xv) Assertion–Reason type:

xi.

A. Only one Barr body is found in normal female

R. Barr body represents the inactivated X-chromosome in somatic cells.

- a) Both A and R are true and R is the reason for A
- b) Both A and R are true and R is not the reason for A
- c) A is correct but R is incorrect
- d) A is incorrect but R is correct

✓ Answer: a) Both A and R are true and R is the reason for A

xii.

A. Frey's syndrome is characterized by sweating over the parotid region during mastication

R. In Frey's syndrome, there is a cross communication between great auricular and auriculotemporal nerves

- a) Both A and R are true and R is the reason for A
- b) Both A and R are true and R is not the reason for A
- c) A is correct but R is incorrect
- d) A is incorrect but R is correct

✅ Answer: c) A is correct but R is incorrect

xiii.

A. Superior vesical artery is the persistent proximal part of the umbilical artery in fetus

R. It is the largest branch of posterior division of internal iliac artery

- a) Both A and R are true and R is the reason for A
- b) Both A and R are true and R is NOT the reason for A
- c) A is correct but R is incorrect
- d) A is incorrect but R is correct

✅ Answer: c) A is correct but R is incorrect

xiv.

A. Greater occipital nerve is a content of suboccipital triangle

R. Greater occipital nerve arises from the dorsal ramus of C2 and travels between the atlas and axis

- a) Both A and R are true and R is the reason for A
- b) Both A and R are true and R is NOT the reason for A
- c) A is correct but R is incorrect

d) A is incorrect but R is correct

✓ Answer: d) A is incorrect but R is correct

xv.

A. Renal colic pain radiates from the renal angle to the groin

R. Posterior surface of lower part of kidney lies deep to the renal angle

a) Both A and R are true and R is the reason for A

b) Both A and R are true and R is NOT the reason for A

c) A is correct but R is incorrect

d) A is incorrect but R is correct

✓ Answer: b) Both A and R are true and R is NOT the reason for A

Question numbers (xvi–xx) Multiple response type:

xvi. Calculi of submandibular gland is common because:

i) Secretions of the gland are mucous & viscid in nature

ii) There is an internal genu for the submandibular duct

iii) The duct drains its secretions against gravity

iv) The secretion is watery

a) i, ii, iii

b) i, iii, iv

c) i, ii, iv

d) iii, iv

✓ Answer: a) i, ii, iii

xvii. Which among the following are the microscopic characteristics of vermiform appendix?

- i) Poorly formed crypts
- ii) Lined by simple squamous epithelium
- iii) Presence of tenia coli
- iv) Abundant lymphoid tissue in submucosa

- a) i, ii, iii
- b) i, iii, iv
- c) i, iv
- d) ii, iii, iv

✓ Answer: c) i, iv

xviii. Which of the following are CORRECT about testis?

- i) Tunica albuginea gives rise to mediastinum testis
- ii) Testicular artery is the branch of anterior division of internal iliac artery
- iii) Hydrocele is the collection of fluid between the layers of tunica vaginalis
- iv) Appendix of testis is the remnant of mesonephric duct

- a) i, iii
- b) i, iii, iv
- c) i, ii
- d) i, iv

✓ Answer: a) i, iii

xix. Which of the following are the actions of superior oblique muscle in primary position of eye ball?

- i) Depression and abduction
- ii) Elevation
- iii) Abduction and extorsion

iv) Intorsion

a) i only

b) i, iv

c) i, iii, iv

d) ii, iii

✓ Answer: b) i, iv

xx. Which of the following are CORRECT about uterus?

i) Develops from paramesonephric duct

ii) Arterial supply from uterine and ovarian arteries

iii) Broad ligament is its main support

iv) Anteversion is the angle between cervix and uterus

a) i, ii

b) i, ii

c) i, ii, iii

d) i, iv

✓ Answer: b) i, ii

Question numbers (i–v) are single response type questions.

i. Which of the following cranial nerve emerges from the posterior aspect of brain stem?

a) spinal accessory

b) vagus

c) hypoglossal

d) trochlear

✓ Answer: d) trochlear

ii. Which of the following cell occupies the Howship's lacunae?

- a) osteoblast
- b) osteocyte
- c) osteoclast
- d) osteoprogenitor cell

✓ Answer: c) osteoclast

iii. Which of the following nerve supplies teres minor?

- a) median
- b) ulnar
- c) musculocutaneous
- d) axillary

✓ Answer: d) axillary

iv. Inferior thyroid artery is a branch of?

- a) arch of aorta
- b) thyrocervical trunk
- c) costocervical trunk
- d) second part of subclavian artery

✓ Answer: b) thyrocervical trunk

v. Which of the following paranasal air sinus develops first?

- a) maxillary
- b) sphenoid
- c) ethmoid

d) frontal

✓ Answer: a) maxillary

Question numbers (vi-x) are case scenario based questions:

A 12 year old boy was brought to the ENT outpatient department with complaints of fever, sore throat, difficulty in swallowing and ear ache for 1 day. On examination it was found that both tonsils were enlarged and diagnosed as acute tonsillitis.

vi. Which of the following lymph node is the principal lymph node of tonsil?

a) jugulodigastric

b) juguloomohyoid

c) submental

d) submandibular

✓ Answer: a) jugulodigastric

vii. Which of the following is the developmental source of tonsil?

a) first pharyngeal pouch

b) second pharyngeal pouch

c) second pharyngeal arch

d) first pharyngeal cleft

✓ Answer: b) second pharyngeal pouch

viii. Which of the following vessel is mainly responsible for bleeding after tonsillectomy?

a) facial artery

b) palatine vein

c) Greater palatine artery

d) paratonsillar vein

✓ Answer: d) paratonsillar vein

ix. Which of the following nerve is responsible for referred pain to the ear in tonsillitis?

a) hypoglossal

b) glossopharyngeal

c) mandibular

✓ Answer: b) glossopharyngeal

x. Which of the following structure is NOT forming the tonsillar bed?

a) pharyngobasilar fascia

b) superior constrictor muscle

c) buccopharyngeal fascia

d) palatoglossus

✓ Answer: d) palatoglossus

Question numbers (xi–xv) Assertion–Reason type:

xi.

A. Cornea is colourless and vascular

R. Cornea is lined by stratified squamous non keratinized epithelium

a) Both A and R are true and R is the reason for A

b) Both A and R are true and R is not the reason for A

c) A is correct but R is incorrect

d) A is incorrect but R is correct

✓ Answer: d) A is incorrect but R is correct

xii.

A. Facial colliculus is seen on the floor of fourth ventricle

R. Motor fibers of facial nerve wind around abducent nerve nucleus

a) Both A and R are true and R is the reason for A

b) Both A and R are true and R is not the reason for A

c) A is correct but R is incorrect

d) A is incorrect but R is correct

✓ Answer: a) Both A and R are true and R is the reason for A

xiii.

A. Skin of the entire upper lip innervated by maxillary nerve

R. Fused globular processes and bilateral maxillary processes form the upper lip

a) Both A and R are true and R is the reason for A

b) Both A and R are true and R is NOT the reason for A

c) A is correct but R is incorrect

d) A is incorrect but R is correct

✓ Answer: b) Both A and R are true and R is NOT the reason for A

xiv.

A. Thyroid gland is removed along with its true capsule during thyroidectomy

R. Venous plexus of thyroid gland lies deep to the true capsule

a) Both A and R are true and R is the reason for A

b) Both A and R are true and R is NOT the reason for A

c) A is correct but R is incorrect

d) A is incorrect but R is correct

✓ Answer: a) Both A and R are true and R is the reason for A

xv.

A. Erb's palsy results from stretching of upper trunk of brachial plexus

R. In Erb's palsy, the arm is adducted & medially rotated, forearm is flexed & supinated in position

- a) Both A and R are true and R is the reason for A
- b) Both A and R are true and R is NOT the reason for A
- c) A is correct but R is incorrect
- d) A is incorrect but R is correct

✓ Answer: a) Both A and R are true and R is the reason for A

Question numbers (xvi–xx) Multiple response type questions:

xvi. The proximal articular surface of wrist joint is formed by,

- i) Lower end of radius
- ii) lower end of ulna
- iii) articular disc of inferior radioulnar joint
- iv) scaphoid, lunate, triquetral

- a) i, ii, iii
- b) iv
- c) i, ii
- d) i, iv

✓ Answer: d) i, iv

xvii. Which among the following muscle(s) depresses the eye in adducted position?

- i) Superior rectus
- ii) Superior oblique
- iii) Inferior oblique
- iv) Inferior rectus

- a) i
- b) ii
- c) i, iii
- d) i, ii

✓ Answer: b) ii (Superior oblique)

xviii. Which of the following statements are TRUE about mammary gland?

- i) It is a modified sebaceous gland
- ii) It extends from 2nd to 6th rib
- iii) It lies within the superficial fascia of the pectoral region
- iv) Involvement of ligaments of Cooper in carcinoma breast results in dimpling of skin

- a) i, ii
- b) iii, iv
- c) ii, iii, iv
- d) i, iv

✓ Answer: c) ii, iii, iv

xix. Which of the following statements are TRUE about falx cerebri?

- i) It fuses posteriorly with tentorium cerebelli
- ii) It separates the cerebellar hemispheres
- iii) It contains the superior sagittal sinus
- iv) It lies in the transverse fissure

- a) i, ii
- b) i, iii
- c) ii, iii
- d) ii, iv

✓ Answer: b) i, iii

xx. Which of the following are the components of striatum?

- i) caudate nucleus
- ii) putamen
- iii) globus pallidus
- iv) lentiform nucleus

- a) i, ii
- b) i, iii
- c) i, iv
- d) i, ii

✓ Answer: d) i, ii

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Question numbers (i–v) are single response type questions.

i. The vertebral level at which the oesophagus pierces the diaphragm is

- a) T6
- b) T8
- c) T10
- d) T12

✓ Answer: c) T10

ii. Which of the following bone violates the law of ossification?

- a) tibia
- b) fibula
- c) talus
- d) calcaneum

✓ Answer: c) talus

iii. Which of the following epithelium lines the endometrium of uterus?

- a) simple squamous
- b) simple cuboidal
- c) pseudostratified columnar
- d) simple columnar

✓ Answer: d) simple columnar

iv. Which of the following is NOT forming the boundary of caudate lobe of liver?

- a) fissure for ligamentum teres
- b) fissure for ligamentum venosum
- c) porta hepatis
- d) groove for inferior vena cava

✓ Answer: a) fissure for ligamentum teres

v. Which is the narrowest part of male urethra?

- a) prostatic
- b) membranous

- c) pre-prostatic
- d) external urethral meatus

✓ Answer: d) external urethral meatus

Question numbers (vi–x) are case scenario based questions:

A 58 year old male presents to the emergency department with epigastric pain radiating to the back, nausea, and vomiting. He has a history of chronic alcohol use. On examination, he is febrile, jaundiced and has tenderness in the epigastric region. Serum amylase and lipase levels are significantly elevated. A contrast-enhanced CT scan shows inflammation of the pancreas, predominantly affecting the head of the pancreas.

vi. Which of the following is the most likely reason for radiation of pain to the back?

- a) irritation of phrenic nerve
- b) irritation of visceral peritoneum
- c) referred pain through inter costal nerves
- d) irritation of parietal peritoneum

✓ Answer: c) referred pain through inter costal nerves

vii. Which of the following structure is compressed in this case leading to jaundice?

- a) right hepatic duct
- b) accessory pancreatic duct
- c) common bile duct
- d) cystic duct

✓ Answer: c) common bile duct

viii. Which part of pancreas is responsible for the production of the enzymes elevated this case?

- a) endocrine part
- b) exocrine part
- c) centroacinar part
- d) both a & b

✓ Answer: b) exocrine part

ix. Where does the main pancreatic duct (duct of Wirsung) drains?

- a) Common bile duct
- b) Jejunum
- c) Duodenum at the minor duodenal papilla
- d) Duodenum at the major duodenal papilla

✓ Answer: d) Duodenum at the major duodenal papilla

x. Which of the following structure develops from ventral pancreatic bud?

- a) Neck
- b) Body
- c) Uncinate process
- d) Tail

✓ Answer: c) Uncinate process

Question numbers (xi–xv) Assertion–Reason type:

xi.

A. Appendicitis may cause periumbilical pain initially.

R. Visceral pain from appendix is referred via T10 dermatome.

- a) Both A and R are true and R is the reason for A

b) Both A and R are true and R is not the reason for A

c) A is correct but R is incorrect

d) A is incorrect but R is correct

✓ Answer: a) Both A and R are true and R is the reason for A

xii.

A. Thoracic duct passes through the posterior mediastinum.

R. Thoracic duct drains into the junction of left internal jugular and left subclavian veins.

a) Both A and R are true and R is the reason for A

b) Both A and R are true and R is not the reason for A

c) A is correct but R is incorrect

d) A is incorrect but R is correct

✓ Answer: b) Both A and R are true and R is not the reason for A

xiii.

A. Great saphenous vein receives small saphenous vein as tributary in all individuals.

R. Small saphenous vein usually drains into popliteal vein, but may occasionally join great saphenous vein.

a) Both A and R are true and R is the reason for A

b) Both A and R are true and R is NOT the reason for A

c) A is correct but R is incorrect

d) A is incorrect but R is correct

✓ Answer: d) A is incorrect but R is correct

xiv.

A. Human kidney develops in 3 successive stages as pronephros, mesonephros and metanephros.

R. All the 3 above mentioned stages are functional in human adults.

- a) Both A and R are true and R is the reason for A
- b) Both A and R are true and R is NOT the reason for A
- c) A is correct but R is incorrect
- d) A is incorrect but R is correct

✔ Answer: c) A is correct but R is incorrect

xv.

A. Ischial spine is an important landmark for administering pudendal nerve block.

R. Pudendal nerve passes close to the ischial spine.

- a) Both A and R are true and R is the reason for A
- b) Both A and R are true and R is NOT the reason for A
- c) A is correct but R is incorrect
- d) A is incorrect but R is correct

✔ Answer: a) Both A and R are true and R is the reason for A

Question numbers (xvi–xx) are multiple response type questions.

xvi. Which of the following are the results of injury to common peroneal nerve?

- i) Foot drop
- ii) loss of plantar flexion
- iii) sensory loss over the dorsum of foot
- iv) wasting of calf muscles

- a) i, ii
- b) i, iii
- c) i, ii, iv

d) i, iii, iv

✓ Answer: b) i, iii

xvii. Which of the following organs are retroperitoneal?

i) Pancreas except tail

ii) kidney

iii) stomach

iv) transverse colon

a) i, ii

b) ii, iii

c) i, ii, iv

d) ii, iii, iv

✓ Answer: a) i, ii

xviii. Which of the following arteries are the branches of internal thoracic artery?

i) Inferior epigastric

ii) musculophrenic

iii) pericardiophrenic

iv) posterior intercostal

a) i, ii

b) i, ii, iv

c) ii, iii

d) ii, iii, iv

✓ Answer: c) ii, iii

xix. Which of the following are the contents of broad ligament of uterus?

- i) Uterine tube
- ii) Round ligament
- iii) Ureter
- iv) Ovarian ligament

- a) i, ii
- b) i, ii, iii
- c) i, ii, iv
- d) all of the above

✔ Answer: d) all of the above

xx. Which of the following are the contents of adductor canal?

- i) femoral artery
- ii) great saphenous vein
- iii) medial cutaneous nerve of thigh
- iv) nerve to vastus medialis

- a) i, ii
- b) i, iii
- c) i, iv
- d) i, ii, iv

✔ Answer: c) i, iv